

- 4 (a) work done per unit positive charge [1]
moving a small test charge from infinity to that point [1]
- (b) (i) $\frac{1}{4\pi\epsilon_0} \frac{2.0 \times 10^{-9}}{4.0 \times 10^{-2}} + \frac{1}{4\pi\epsilon_0} \frac{Q}{8.0 \times 10^{-2}} = 0$ [1]
Solving, $Q = -4.0 \times 10^{-9} \text{ C}$ [1]
- (ii) Loss in KE = Gain in EPE
 $\frac{1}{2} mv^2 = q\Delta V$
 $\frac{1}{2} (4 \times 1.66 \times 10^{-27})(v^2) = (2 \times 1.6 \times 10^{-19})(200 - (-1000))$ [1]
Solving, $v = 3.4 \times 10^5 \text{ m s}^{-1}$ [1]
- (c) (i) a region of space where a charge experiences an electric force [1]
- (ii) similarity – any one point from:
 - both have an inverse square function
 - both decrease with distance
 - both are radial [1]
difference – any one point from:
 - gravitational field is always towards the mass
 - electric field can be towards or away from the charge [1]
- (d) (i) $\frac{1}{4\pi\epsilon_0} \frac{Q}{0.15^2} = 84$
Solving, $Q = 2.1 \times 10^{-10} \text{ C}$ [1]
- (ii) 0 N C^{-1} (electric field strength = 0 N C^{-1} inside a metal sphere) [1]

